

Definitions in **Sarason/** and **R2.lean**

Lean Formalization Project

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This document provides a comprehensive list of all definitions manually formalized in the **ComplexAnalysis** project, specifically within **R2.lean** and the **Sarason/** directory. For each definition, the raw Lean code is presented first, followed by its mathematical interpretation.

1 **ComplexAnalysis/R2.lean**

- **sqNorm** ($x \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The squared Euclidean norm.

```
noncomputable def sqNorm (x : ℝ × ℝ) : ℝ := x.12 + x.22
```

$$\text{sqNorm}(x) = x_1^2 + x_2^2$$

- **sqDist** ($x, y \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The squared Euclidean distance.

```
noncomputable def sqDist (x y : ℝ × ℝ) : ℝ := sqNorm (x - y)
```

$$\text{sqDist}(x, y) = \text{sqNorm}(x - y)$$

- **euclideanNorm** ($x \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The standard Euclidean norm.

```
noncomputable def euclideanNorm (x : ℝ × ℝ) : ℝ := Real.sqrt (sqNorm x)
```

$$\text{euclideanNorm}(x) = \sqrt{x_1^2 + x_2^2}$$

- **euclideanDist** ($x, y \in \mathbb{R}^2 \rightarrow \mathbb{R}$): The Euclidean distance.

```
noncomputable def euclideanDist (x y : ℝ × ℝ) : ℝ := Real.sqrt (sqDist x y)
```

$$\text{euclideanDist}(x, y) = \sqrt{\text{sqDist}(x, y)}$$

- **LimitRtoR2** ($f : \mathbb{R} \rightarrow \mathbb{R}^2, a \in \mathbb{R}, L \in \mathbb{R}^2$):

```
def LimitRtoR2 (f : ℝ → ℝ × ℝ) (a : ℝ) (L : ℝ × ℝ) : Prop :=  
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ, 0 < |x - a| ∧ |x - a| < δ → euclideanDist (f x) L < ε
```

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0, \forall x, 0 < |x - a| < \delta \implies \text{euclideanDist}(f(x), L) < \epsilon$$

- **LimitR2toR** ($f : \mathbb{R}^2 \rightarrow \mathbb{R}, a \in \mathbb{R}^2, L \in \mathbb{R}$):

```
def LimitR2toR (f : ℝ × ℝ → ℝ) (a : ℝ × ℝ) (L : ℝ) : Prop :=  
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ × ℝ, 0 < euclideanDist x a ∧ euclideanDist x a < δ → |  
    f x - L| < ε
```

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}^2, 0 < \text{euclideanDist}(x, a) < \delta \implies |f(x) - L| < \epsilon$$

- **LimitR2toR2** ($f : \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $a \in \mathbb{R}^2$, $L \in \mathbb{R}^2$):

```
def LimitR2toR2 (f : ℝ × ℝ → ℝ × ℝ) (a : ℝ × ℝ) (L : ℝ × ℝ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ × ℝ, 0 < euclideanDist x a ∧ euclideanDist x a < δ →
    euclideanDist (f x) L < ε
```

$$\lim_{x \rightarrow a} f(x) = L \iff \forall \epsilon > 0, \exists \delta > 0, \forall x \in \mathbb{R}^2, 0 < \text{euclideanDist}(x, a) < \delta \implies \text{euclideanDist}(f(x), L) < \epsilon$$

- **ConvergesR2** ($\text{seq} : \mathbb{N} \rightarrow \mathbb{R}^2$, $L \in \mathbb{R}^2$):

```
def ConvergesR2 (seq : ℕ → ℝ × ℝ) (L : ℝ × ℝ) : Prop :=
  ∀ ε > 0, ∃ N : ℕ, ∀ n ≥ N, euclideanDist (seq n) L < ε
```

$$\lim_{n \rightarrow \infty} \text{seq}(n) = L \iff \forall \epsilon > 0, \exists N \in \mathbb{N}, \forall n \geq N, \text{euclideanDist}(\text{seq}(n), L) < \epsilon$$

- **HasFDerivAt_R2_eps** ($u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$, $u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}$, $a \in \mathbb{R}^2$): Defines the Fréchet derivative of $F(x, y) = (u(x, y), v(x, y))$ at a . Let $L(h) = \begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} h$.

```
def HasFDerivAt_R2_eps (u v : ℝ × ℝ → ℝ) (ux0 uy0 vx0 vy0 : ℝ) (a : ℝ × ℝ) :
  Prop :=
  LimitR2toR2 (fun h => (
    (u (a.1 + h.1, a.2 + h.2) - u a) - (ux0 * h.1 + uy0 * h.2),
    (v (a.1 + h.1, a.2 + h.2) - v a) - (vx0 * h.1 + vy0 * h.2)
  )) (0, 0) (0, 0)
```

$$\lim_{h \rightarrow 0} \left(F(a + h) - F(a) - \begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} h \right) = 0$$

$$\lim_{(h_1, h_2) \rightarrow (0, 0)} \left(\begin{pmatrix} u(a_1 + h_1, a_2 + h_2) - u(a_1, a_2) \\ v(a_1 + h_1, a_2 + h_2) - v(a_1, a_2) \end{pmatrix} - \begin{pmatrix} u_{x0} & u_{y0} \\ v_{x0} & v_{y0} \end{pmatrix} \begin{pmatrix} h_1 \\ h_2 \end{pmatrix} \right) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

- **DifferentiableAt_R2_eps** ($u, v : \mathbb{R}^2 \rightarrow \mathbb{R}$, $a \in \mathbb{R}^2$):

```
def DifferentiableAt_R2_eps (u v : ℝ × ℝ → ℝ) (a : ℝ × ℝ) : Prop :=
  ∃ ux0 uy0 vx0 vy0 : ℝ, HasFDerivAt_R2_eps u v ux0 uy0 vx0 vy0 a
```

$$\exists u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}, \text{HasFDerivAt_R2_eps}(u, v, u_{x0}, u_{y0}, v_{x0}, v_{y0}, a)$$

- **HasDerivAt_RtoR2_eps** ($f : \mathbb{R} \rightarrow \mathbb{R}^2$, $f'_0 \in \mathbb{R}^2$, $t_0 \in \mathbb{R}$): Defines the derivative of a parametric curve in $\mathbb{R}^2$.

```
def HasDerivAt_RtoR2_eps (f : ℝ → ℝ × ℝ) (f'0 : ℝ × ℝ) (t0 : ℝ) : Prop :=
  LimitRtoR2 (fun t => (
    (f t).1 - (f t0).1 - (t - t0) * f'0.1,
    (f t).2 - (f t0).2 - (t - t0) * f'0.2
  )) t0 (0, 0)
```

$$\lim_{t \rightarrow t_0} (f(t) - f(t_0) - (t - t_0)f'_0) = 0$$

2 ComplexAnalysis/Sarason/Definitions.lean

- **HasLimitAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, L \in \mathbb{C}, z_0 \in \mathbb{C}$):

```
def HasLimitAt_eps (f : ℂ → ℂ) (L : ℂ) (z₀ : ℂ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ z : ℂ, 0 < Complex.abs (z - z₀) ∧ Complex.abs (z - z₀) < δ →
    Complex.abs (f z - L) < ε
```

$$\lim_{z \rightarrow z_0} f(z) = L$$

- **HasLimitAtInf_eps** ($f : \mathbb{C} \rightarrow \mathbb{R}, L \in \mathbb{R}$):

```
def HasLimitAtInf_eps (f : ℂ → ℝ) (L : ℝ) : Prop :=
  ∀ ε > 0, ∃ R > 0, ∀ z : ℂ, Complex.abs z > R → |f z - L| < ε
```

$$\lim_{|z| \rightarrow \infty} f(z) = L$$

- **HasDerivAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, f'_0 \in \mathbb{C}, z_0 \in \mathbb{C}$): Complex differentiability at a point.

```
def HasDerivAt_eps (f : ℂ → ℂ) (f'₀ : ℂ) (z₀ : ℂ) : Prop :=
  HasLimitAt_eps (fun z => (f z - f z₀) / (z - z₀)) f'₀ z₀
```

$$\lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0} = f'_0$$

- **DifferentiableAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$):

```
def DifferentiableAt_eps (f : ℂ → ℂ) (z₀ : ℂ) : Prop :=
  ∃ f'₀, HasDerivAt_eps f f'₀ z₀
```

f is differentiable at z_0

- **HasPartialDerivX_C_to_R_eps** / **HasPartialDerivY_C_to_R_eps**: Partial derivatives of a real-valued function of a complex variable (identifying $\mathbb{C} \cong \mathbb{R}^2$).

```
def HasPartialDerivX_C_to_R_eps (u : ℂ → ℝ) (ux : ℝ) (z₀ : ℂ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ x : ℝ, 0 < |x - z₀.re| ∧ |x - z₀.re| < δ →
    |(u ((x : ℂ) + (z₀.im : ℂ) * I) - u z₀) / (x - z₀.re) - ux| < ε
```

$$\frac{\partial u}{\partial x}(z_0) = u_x$$

- **HasPartialDerivY_C_to_R_eps**

```
def HasPartialDerivY_C_to_R_eps (u : ℂ → ℝ) (uy : ℝ) (z₀ : ℂ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ y : ℝ, 0 < |y - z₀.im| ∧ |y - z₀.im| < δ →
    |(u ((z₀.re : ℂ) + (y : ℂ) * I) - u z₀) / (y - z₀.im) - uy| < ε
```

$$\frac{\partial u}{\partial y}(z_0) = u_y$$

- **deriv_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}, h : \text{DifferentiableAt_eps } f \ z_0$): The classical derivative value $f'(z_0)$ extracted via choice.

```
noncomputable def deriv_eps (f : ℂ → ℂ) (z₀ : ℂ) (h : DifferentiableAt_eps f z₀) : ℂ :=
  Classical.choose h
```

$$\text{deriv_eps}(f, z_0) = f'(z_0)$$

- **HasDerivAt_R_to_C_eps** ($f : \mathbb{R} \rightarrow \mathbb{C}$, $f'_0 \in \mathbb{C}$, $t_0 \in \mathbb{R}$): Derivative of a complex-valued path parameterized by a real variable.

```
def HasDerivAt_R_to_C_eps (f : ℝ → ℂ) (f'_0 : ℂ) (t_0 : ℝ) : Prop :=
  ∀ ε > 0, ∃ δ > 0, ∀ t : ℝ, 0 < |t - t_0| ∧ |t - t_0| < δ →
    Complex.abs ((f t - f t_0) / (t - t_0) - f'_0) < ε
```

$$\lim_{t \rightarrow t_0} \frac{f(t) - f(t_0)}{t - t_0} = f'_0$$

- **HolomorphicAt_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}$, $z_0 \in \mathbb{C}$): f is complex-differentiable in some open neighborhood around z_0 .

```
def HolomorphicAt_eps (f : ℂ → ℂ) (z_0 : ℂ) : Prop :=
  ∃ r > 0, ∀ z, Complex.abs (z - z_0) < r → DifferentiableAt_eps f z
```

f is holomorphic at $z_0 \iff f$ is complex-differentiable on some open ball around z_0

- **HolomorphicOn_eps** ($f : \mathbb{C} \rightarrow \mathbb{C}$, $G \subset \mathbb{C}$): f is holomorphic at every point in the open set G .

```
def HolomorphicOn_eps (f : ℂ → ℂ) (G : Set ℂ) : Prop :=
  ∀ z ∈ G, HolomorphicAt_eps f z
```

f is holomorphic on G

3 ComplexAnalysis/Sarason/Chapter1.lean

- **chordalMetric** ($z, w \in \mathbb{C} \rightarrow \mathbb{R}$): The chordal distance between two points in the complex plane mapped to the Riemann sphere.

```
noncomputable def chordalMetric (z w : ℂ) : ℝ :=
  (2 * Complex.abs (z - w)) / (Real.sqrt (1 + Complex.normSq z) * Real.sqrt (1
    + Complex.normSq w))
```

$$\chi(z, w) = \frac{2|z - w|}{\sqrt{1 + |z|^2} \sqrt{1 + |w|^2}}$$

- **chordalMetricToInf** ($z \in \mathbb{C} \rightarrow \mathbb{R}$): The chordal distance from a point z to the point at infinity ∞ .

```
noncomputable def chordalMetricToInf (z : ℂ) : ℝ :=
  2 / Real.sqrt (1 + Complex.normSq z)
```

$$\chi(z, \infty) = \frac{2}{\sqrt{1 + |z|^2}}$$

- **Φ (Stereographic Projection)**

```
noncomputable def Φ (z : ℂ) : EuclideanSpace ℝ (Fin 3) :=
  ![ (2 * z.re) / (1 + Complex.normSq z),
    (2 * z.im) / (1 + Complex.normSq z),
    (Complex.normSq z - 1) / (1 + Complex.normSq z) ]
```

$$\Phi(z) = \left(\frac{2\operatorname{Re}(z)}{1 + |z|^2}, \frac{2\operatorname{Im}(z)}{1 + |z|^2}, \frac{|z|^2 - 1}{1 + |z|^2} \right) \in S^2$$

- **ChordalC**: Type alias for $\mathbb{C}$ equipped with the chordal metric.

```
def ChordalC : Type := ℂ
```

$\text{ChordalC} = \mathbb{C}$ (equipped with chordal metric)

4 ComplexAnalysis/Sarason/Chapter2.lean

- `del` (∂)

```
def del (f : ℂ → ℂ) (z : ℂ) : ℂ :=
  (1/2 : ℂ) * (deriv (fun x : ℝ => f (z + x)) 0 - I * deriv (fun y : ℝ => f (z
    + y * I)) 0)
```

$$\partial f = \frac{1}{2} \left(\frac{\partial f}{\partial x} - i \frac{\partial f}{\partial y} \right)$$

- `delBar` ($\bar{\partial}$)

```
def delBar (f : ℂ → ℂ) (z : ℂ) : ℂ :=
  (1/2 : ℂ) * (deriv (fun x : ℝ => f (z + x)) 0 + I * deriv (fun y : ℝ => f (z
    + y * I)) 0)
```

$$\bar{\partial} f = \frac{1}{2} \left(\frac{\partial f}{\partial x} + i \frac{\partial f}{\partial y} \right)$$

- `del_eps` / `delBar_eps` ($u_{x0}, u_{y0}, v_{x0}, v_{y0} \in \mathbb{R}$): Wirtinger derivatives expressed purely in terms of the real partial derivatives of $f = u + iv$.

```
noncomputable def del_eps (ux0 uy0 vx0 vy0 : ℝ) : ℂ :=
  ((ux0 + vy0) / 2 : ℝ) + I * ((vx0 - uy0) / 2 : ℝ)
```

$$\partial f = \frac{1}{2}(u_{x0} + v_{y0}) + \frac{i}{2}(v_{x0} - u_{y0})$$

- `delBar_eps`

```
noncomputable def delBar_eps (ux0 uy0 vx0 vy0 : ℝ) : ℂ :=
  ((ux0 - vy0) / 2 : ℝ) + I * ((vx0 + uy0) / 2 : ℝ)
```

$$\bar{\partial} f = \frac{1}{2}(u_{x0} - v_{y0}) + \frac{i}{2}(v_{x0} + u_{y0})$$

- `path_in_C` ($x, y \in \mathbb{C}$): A continuous path $\gamma : [0, 1] \rightarrow \mathbb{C}$ starting at x and ending at y .

```
def path_in_C (x y : ℂ) : Type := Path x y
```

$$\gamma : [0, 1] \rightarrow \mathbb{C} \text{ where } \gamma(0) = x, \gamma(1) = y$$

- `translated_path` ($c, x, y \in \mathbb{C}, \gamma : \text{path_in_C } x \ y$): A path shifted by the constant c , creating a path from $c + x$ to $c + y$.

```
noncomputable def translated_path (c x y : ℂ) (γ : path_in_C x y) : path_in_C
  (c + x) (c + y) where
  toFun := fun t => c + γ t
```

$$\gamma_c(t) = c + \gamma(t)$$

- `pathDeriv` ($\gamma : \text{path_in_C } x \ y, t \in \mathbb{R}$): The tangent vector $\gamma'(t) \in \mathbb{C}$, extracted via choice from `HasDerivAt_R_to_C_eps` extended to $\mathbb{R}$.

```
noncomputable def pathDeriv {x y : ℂ} (γ : path_in_C x y) (t : ℝ) : ℂ :=
  deriv γ.extend t
```

$$\gamma'(t) = \frac{d\gamma}{dt}$$

- `pathDirection` ($\gamma : \text{path_in_C } x \ y, t \in \mathbb{R}$): The argument (angle) of the tangent vector, $\arg(\gamma'(t))$.

```
noncomputable def pathDirection {x y : ℂ} (γ : path_in_C x y) (t : ℝ) : ℝ :=
  Complex.arg (pathDeriv γ t)
```

$$\text{Direction}(\gamma, t) = \arg(\gamma'(t))$$

- `pathAngle` ($\gamma_1, t_1, \gamma_2, t_2$): The angle between two intersecting paths.

```
noncomputable def pathAngle {x1 y1 x2 y2 : ℂ} (γ1 : path_in_C x1 y1) (t1 : ℝ)
  (γ2 : path_in_C x2 y2) (t2 : ℝ) : ℝ :=
  pathDirection γ2 t2 - pathDirection γ1 t1
```

$$\text{Angle}(\gamma_1, t_1, \gamma_2, t_2) = \arg(\gamma_2'(t_2)) - \arg(\gamma_1'(t_1))$$

- `conformal` ($f : \mathbb{C} \rightarrow \mathbb{C}, z_0 \in \mathbb{C}$): f preserves angles between any two paths intersecting at z_0 with non-zero derivatives.

```
def conformal (f : ℂ → ℂ) (z0 : ℂ) : Prop :=
  ∀ x1 y1 x2 y2 : ℂ, ∀ γ1 : path_in_C x1 y1, ∀ t1 : ℝ,
  ∀ γ2 : path_in_C x2 y2, ∀ t2 : ℝ,
  γ1.extend t1 = z0 → γ2.extend t2 = z0 →
  DifferentiableAt ℝ γ1.extend t1 → DifferentiableAt ℝ γ2.extend t2 →
  pathDeriv γ1 t1 ≠ 0 → pathDeriv γ2 t2 ≠ 0 →
  DifferentiableAt ℝ (f ∘ γ1.extend) t1 → DifferentiableAt ℝ (f ∘ γ2.extend) t2 →
  pathDeriv (f_circ_gamma γ1 f) t1 ≠ 0 ∧
  pathDeriv (f_circ_gamma γ2 f) t2 ≠ 0 ∧
  pathAngle (f_circ_gamma γ1 f) t1 (f_circ_gamma γ2 f) t2 = pathAngle γ1 t1 γ2 t2
```

Conformal f at $z_0 \iff f$ preserves angles between all regular intersecting curves at z_0

- `centered_line_path` ($z, v \in \mathbb{C}$): A straight-line path from $z - v$ to $z + v$ evaluating to z at $t = 1/2$.

```
noncomputable def centered_line_path (z v : ℂ) : path_in_C (z - v) (z + v)
  where
  toFun := fun t => z + (2 * (t : ℝ) - 1) * v
```

$$\gamma(t) = z + (2t - 1)v$$

5 ComplexAnalysis/Sarason/Chapter3.lean

- `translationMatrix` ($b \in \mathbb{C}$):

```
noncomputable def translationMatrix (b : ℂ) : GL (Fin 2) ℂ :=
  Units.mk0 !![1, b; 0, 1] ...
```

$$\begin{pmatrix} 1 & b \\ 0 & 1 \end{pmatrix}$$

- `scalingMatrix` ($a \in \mathbb{C}, a \neq 0$):

```
noncomputable def scalingMatrix (a : ℂ) (ha : a ≠ 0) : GL (Fin 2) ℂ :=
  Units.mk0 !![a, 0; 0, 1] ...
```

$$\begin{pmatrix} a & 0 \\ 0 & 1 \end{pmatrix}$$

- `inversionMatrix`:

```
noncomputable def inversionMatrix : GL (Fin 2) ℂ :=
  Units.mk0 ![0, 1; 1, 0] ...
```

$$\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$